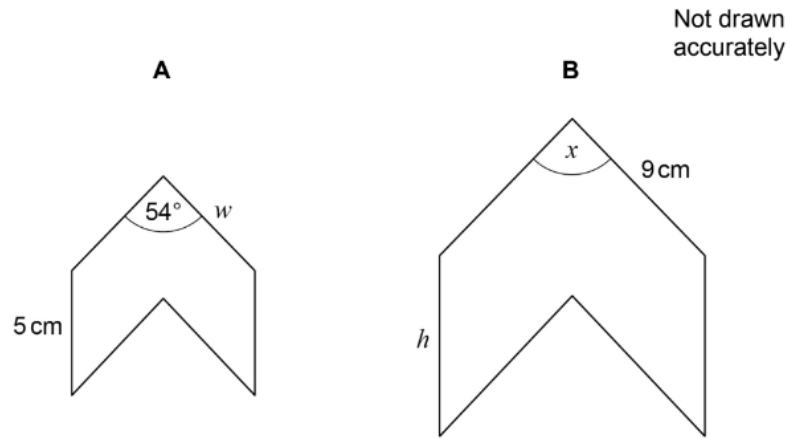


1. Shapes **A** and **B** are similar with scale factor 1.5. Determine the values of h , w , x .



[3 marks]

2. (a) Write 16,200,000 in standard form.

(b) Write 3.62×10^{-5} as an ordinary number.

[2 marks]

3. (a) Determine the mean, median and mode of the following integers:

15, 17, -12, 4, 5, 12, -4, -4.

(b) The mean of 5 numbers is 72. A new number is added and the mean changes to 70. What number was added?

[6 marks]

4. Fully factorise the following expressions:

(a) $x^2 - 11x - 42$

(b) $x^2 - 16x$

(c) $4x^2 - 49$

(d) $12 - x^2 - x$

[5 marks]

5. Calculate the following percentage changes.

(a) Increase 56.2 by 13%.

(b) Decrease 17.4 by 7%.

(c) £500 is invested into a savings account which accrues 1.3% compound interest annually. How much money is in the account after 5 years?

(d) Increase 154 by 7%, then decrease it by 2% and then increase it by 1%. What are you left with?

[5 marks]

6. Calculate the following without the use of a calculator. You must show all stages in your working. A correct answer with no method will not obtain credit.

(a) $(3.2 \times 10^4) \div (4 \times 10^2)$

(b) $(5.21 \times 10^{-3}) - (1.3 \times 10^{-4})$

[4 marks]

7. Solve the following equations.

(a) $x^2 + 5x - 14 = 0$

(b) $(3x - 1)(x + 5) = 0$

(c) $2(x - 1)^2 = x(x - 7)$

[5 marks]

8. (a) The following table summarises how late trains were leaving station **A**. Calculate the mean number of minutes late.

Number of minutes late, t	Number of trains
0	5
1	7
2	12
3	25
4	10
5	31

- (b) The following table summarises how late trains were leaving station **B**. Calculate an estimate for the mean number of minutes late.

Number of minutes late, t	Number of trains
$0 \leq t < 5$	12
$5 \leq t < 10$	7
$10 \leq t < 15$	1
$t \geq 15$	0

- (c) Make two comparisons of the data between station **A** and station **B**.

[6 marks]

9. (a) The price of an item of clothing with 18% off is £65.60. What was the original cost of the item?

- (b) The price of a cup of coffee is increased by 15% and then later decreased by 20%. Express these changes in price as a single percentage change.

[4 marks]

10. Fully simplify the following algebraic fraction.

$$\frac{6a^2 + 6a - 72}{3a^2 - 48}$$

[3 marks]

11. (a) Simplify the following ensuring that your final answers are given in standard form. You may assume that $p, q > 0$.

i. $(p \times 10^m) \times (q \times 10^n)$ where $10 < pq < 100$.

ii. $(p \times 10^{2m+1}) \div (q \times 10^{m-1})$ where $\frac{q}{100} < p < \frac{q}{10}$.

- (b) The number $\frac{2x+1}{10} \times 10^4$ is in standard form. Determine the possible values of x .

[6 marks]

Bonus Question: This question will **NOT** contribute to your final mark.

Solve the following equation for x . You should give your answer in terms of y .

$$\frac{7x^3 - 63xy^2}{xy - 6y^2 + x^2} = 0$$

[4 marks]